

H9

```
library(tidyverse)

## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
## v dplyr      1.2.0      v readr      2.1.6
## v forcats    1.0.1      v stringr   1.6.0
## v ggplot2    4.0.2      v tibble    3.3.1
## v lubridate  1.9.4      v tidyr     1.3.2
## v purrr      1.2.1
## -- Conflicts ----- tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()     masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors

raw_dataset <- read_csv(
  "sentence_memorability_data.csv",
  na = c("", "NA", "N/A"),
  show_col_types = FALSE
)

h9_data <- raw_dataset %>%
  # Make sure RTs are treated as numbers
  mutate(
    Reaction_time_IR = suppressWarnings(as.numeric(Reaction_time_IR)),
    Reaction_time_WR = suppressWarnings(as.numeric(Reaction_time_WR)),
    Accuracy_IR = suppressWarnings(as.numeric(Accuracy_IR)),
    Accuracy_WR = suppressWarnings(as.numeric(Accuracy_WR)),
    prev_IR = lag(Accuracy_IR)
  ) %>%
  group_by(participant_ID) %>%
  summarise(
    # Calculate median reaction times (ignores outliers)
    Median_RT_IR = median(Reaction_time_IR, na.rm = TRUE),
    Median_RT_WR = median(Reaction_time_WR, na.rm = TRUE),

    # IR components
    Total_Repeats = sum(isRepeat == TRUE & Event == "Sentence shown", na.rm = TRUE),
    Total_Denom = sum(is.na(isRepeat) & Event == "Sentence shown", na.rm = TRUE),
    Hits = sum(Event == "IR pressed" & isRepeat == TRUE & Accuracy_IR == 1, na.rm = TRUE),
    False_Alarms = sum(Event == "IR pressed" & is.na(isRepeat), na.rm = TRUE),

    # IR metrics
    Hit_Rate = if_else(Total_Repeats > 0, Hits / Total_Repeats, 0),
    False_Alarm_Rate = if_else(Total_Denom > 0, False_Alarms / Total_Denom, 0),
    IR_Memorability = Hit_Rate - False_Alarm_Rate,

    # WR components
    WR_correct = sum(Event == "WR pressed" & Accuracy_WR == 1 & prev_IR == 1, na.rm = TRUE),
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WR_total = sum(Event == "WR pressed" & prev_IR == 1, na.rm = TRUE),

# WR metric
WR_Accuracy = if_else(WR_total > 0, WR_correct / WR_total, 0),
.groups = "drop"
) %>%
# Drop any participants missing RTs or scores to ensure a clean correlation
drop_na(Median_RT_IR, Median_RT_WR, IR_Memorability, WR_Accuracy)

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library(ggplot2)
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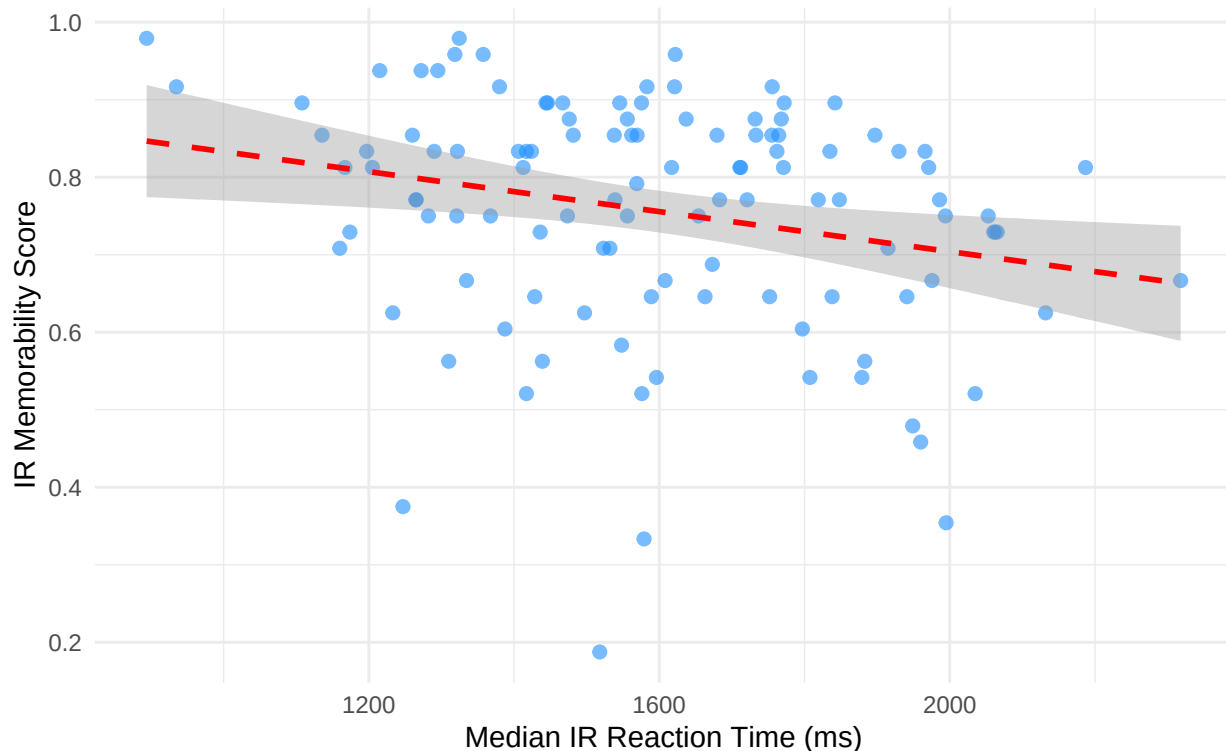
# Graph H9a: IR Reaction Time vs. IR Memorability
ggplot(h9_data, aes(x = Median_RT_IR, y = IR_Memorability)) +
  geom_point(alpha = 0.6, color = "dodgerblue", size = 2) +
  geom_smooth(method = "lm", color = "red", linetype = "dashed", se = TRUE) +
  labs(title = "H9a: IR Reaction Time vs. IR Memorability",
       subtitle = "Testing if faster responses yield lower memorability scores",
       x = "Median IR Reaction Time (ms)",
       y = "IR Memorability Score") +
  theme_minimal()

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```
## `geom_smooth()` using formula = 'y ~ x'
```

H9a: IR Reaction Time vs. IR Memorability

Testing if faster responses yield lower memorability scores



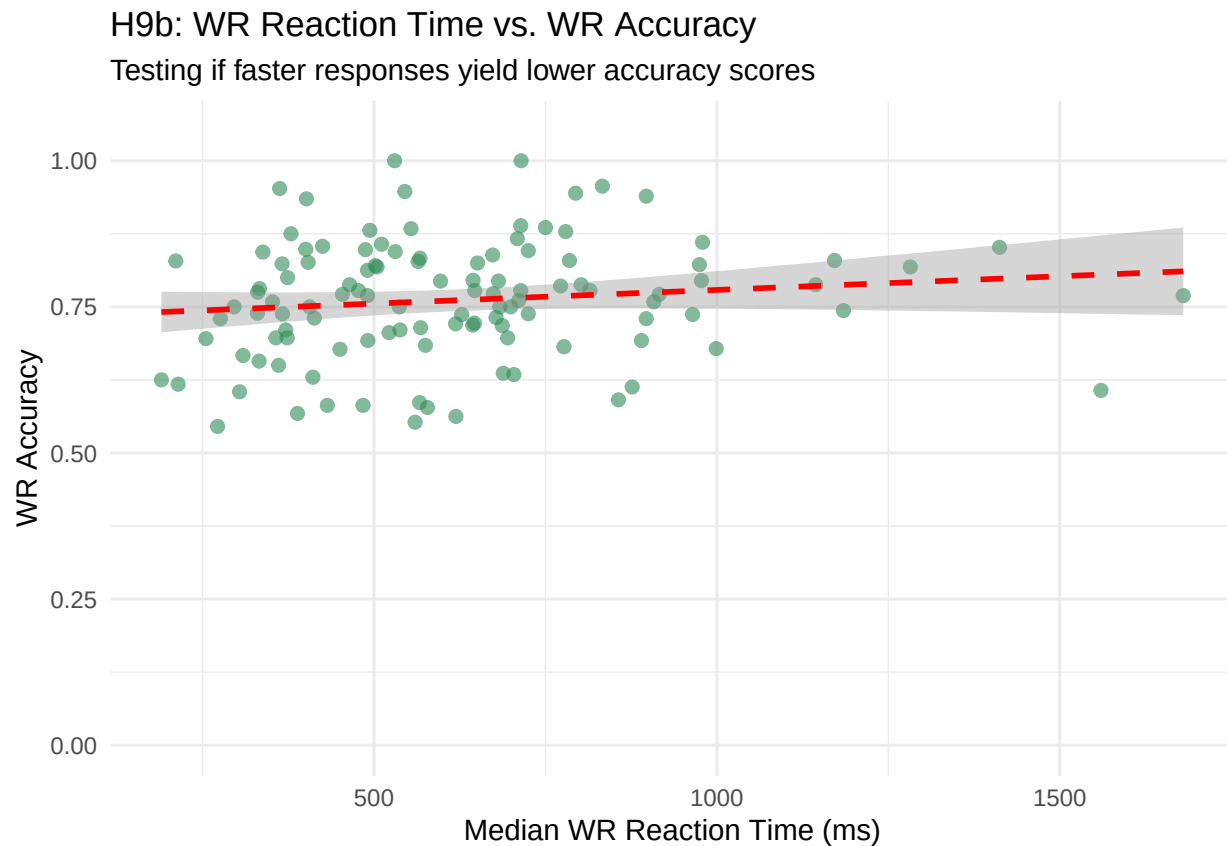
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# Graph H9b: WR Reaction Time vs. WR Accuracy
ggplot(h9_data, aes(x = Median_RT_WR, y = WR_Accuracy)) +
  geom_point(alpha = 0.6, color = "seagreen", size = 2) +
  geom_smooth(method = "lm", color = "red", linetype = "dashed", se = TRUE) +

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scale_y_continuous(limits = c(0, 1.05)) +
labs(title = "H9b: WR Reaction Time vs. WR Accuracy",
      subtitle = "Testing if faster responses yield lower accuracy scores",
      x = "Median WR Reaction Time (ms)",
      y = "WR Accuracy") +
theme_minimal()
```

```
## `geom_smooth()` using formula = 'y ~ x'
```



```
# Spearman Correlation for H9a (IR)
cor_ir <- cor.test(h9_data$Median_RT_IR, h9_data$IR_Memorability, method = "spearman", exact = FALSE)
print("Spearman Correlation for H9a (IR):")
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## [1] "Spearman Correlation for H9a (IR):"
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print(cor_ir)
```

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##
## Spearman's rank correlation rho
##
## data: h9_data$Median_RT_IR and h9_data$IR_Memorability
## S = 310989, p-value = 0.005293
## alternative hypothesis: true rho is not equal to 0
## sample estimates:
## rho
## -0.2595479
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# Spearman Correlation for H9b (WR)
cor_wr <- cor.test(h9_data$Median_RT_WR, h9_data$WR_Accuracy, method = "spearman", exact = FALSE)
print("Spearman Correlation for H9b (WR):")

## [1] "Spearman Correlation for H9b (WR):"
print(cor_wr)

##
## Spearman's rank correlation rho
##
## data: h9_data$Median_RT_WR and h9_data$WR_Accuracy
## S = 203164, p-value = 0.05935
## alternative hypothesis: true rho is not equal to 0
## sample estimates:
## rho
## 0.1771567

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